

Incorrect Speed Pickup Signal Used to Shut Down Unit

Symptoms

- The SDU410 unit engine speed value is incorrect.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SDU410 unit has two engine speed input signals. If the signals differ, the SDU410 unit uses the higher of the two readings as the signal.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check customer interface box	
	STEP 1A. Check the customer interface box logic unit LED illumination.	Are any alarms active or LEDs illuminated?
	STEP 1B. Check the SDU410 power supply wire for +24-VDC.	Less than +24-VDC?
STEP 2.	Check customer interface box wiring	
	STEP 2A. Check the engine speed 1 and engine speed 2 signal and return wires for an open.	Less than 10 ohms?
	STEP 1A-1. Check the engine speed 2 signal and return wires for an open.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 2B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.	Less than 10 ohms?
	STEP 1B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.	Less than 10 ohms?
	STEP 2C. Check the engine speed 1 signal wire for a short to ground at the SDU410 unit and C4 connector.	Less than 10 ohms?
	STEP 2C-1. Check the engine speed 2 signal wire for a short to ground at the SDU410 unit and C4 connector.	Less than 10 ohms?
STEP 3.	Check the OEM wiring harness	
	STEP 3A. Check the engine speed 1 signal and return wires for an open at the C4 and C11 connectors.	Less than 10 ohms?
	STEP 3A-1. Check the engine speed 2 signal and return wires for an open at the C4 and C11 connectors.	Less than 10 ohms?
	STEP 3B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.	Less than 10 ohms?
	STEP 3B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 3C. Check the engine speed 1 signal wire for a short to ground at the C4 connector.	Less than 10 ohms?
	STEP 3C-1. Check the engine speed 2 signal wire for a short to ground at the C4 and C11 connectors.	Less than 10 ohms?

STEP 1. Check customer interface box

STEP 1A. Check the customer interface box logic unit LED illumination.

Conditions :

- Check the DCU410 unit for alarms and LED illumination.

Action	Specification/Repair	Next Step
Check for alarms and LED illumination on the DCU410 unit.	Are any alarms active or LEDs illuminated? YES	Contact a Cummins® Authorized Repair Location
	Are any alarms active or LEDs illuminated? NO	1B

STEP 1B. Check the DCU410 power supply wire for +24-VDC.

Conditions :

- Open the customer interface box

Action	Specification/Repair	Next Step
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Check the voltage at the shutdown unit supply 24-VDC at the SDU410 unit. - Place one test lead on the shutdown unit supply 24-VDC supply wire at the SDU410 unit. - Place the other test lead on the shutdown unit return wire at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete
	Less than +24-VDC? NO	2A

STEP 2. Check customer interface box wiring

STEP 2A. Check the engine speed 1 signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 and engine speed 2 signal and return wires from the SDU410 unit and disconnect the C4 connector.

Action	Specification/Repair	Next Step

Check the signal and return wires for an open. - Place one test lead on the engine speed 1 supply wire at the SDU410 unit. - Place the other test lead on the engine speed 1 signal pin at the C4 connector. - Place one test lead on the engine speed 1 return wire at the SDU410 unit. - Place the other test lead on the engine 1 return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2A-1
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 2A-1. Check the engine speed 2 signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 2 signal and return wires from the SDU410 unit and disconnect the C4 connector.

Action	Specification/Repair	Next Step

Check the signal and return wires for an open. - Place one test lead on the engine speed 2 supply wire at the SDU410 unit. - Place the other test lead on the engine speed 2 signal pin at the C4 connector. - Place one test lead on the engine speed 2 return wire at the SDU410 unit. - Place the other test lead on the engine speed 2 return pin at the C4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 2B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.

Conditions : - Open the customer interface box. - Disconnect the engine speed 1 signal and return wires from the SDU410 unit and disconnect the C4 connector.		
Action	Specification/Repair	Next Step

Check the signal and return wires for a wire-to-wire short. - Place one test lead on the engine speed 1 signal at the SDU410 unit. - Place the other test lead on all other wires on the SDU410 unit. - Place one test lead on the engine speed 1 return wire at the SDU410 unit. - Place the other test lead on all other wires on the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	2B-1

STEP 2B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the SDU410 unit and C4 connector.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed sensor 2 signal and return wires from the SDU410 unit and disconnect the C4 connector.

Action	Specification/Repair	Next Step
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Check the signal and return wires for a wire-to-wire short. - Place one test lead on the engine speed 2 supply wire at the SDU410 unit. - Place the other test lead on all other wires at the SDU410 unit. - Place one test lead on the engine speed 2 return wire at the SDU410 unit. - Place the other test lead on all other wires at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the engine speed 1 signal wire for a short to ground at the SDU410 unit and C4 connector.

Conditions :

- Open the customer interface box.
- Disconnect the engine speed 1 signal wire from the SDU410 unit and disconnect the C4 connector.

Action	Specification/Repair	Next Step
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Check the signal wire for short to ground. - Place one test lead on the engine speed 1 signal wire at the SDU410 unit. - Place the other test lead to panel ground. - Place one test lead on the engine speed 1 signal pin at the C4 connector. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	2C-1

STEP 2C-1. Check the engine speed 2 signal wire for a short to ground at the SDU410 unit and C4 connector.

Conditions : - Open the customer interface box. - Disconnect the engine speed 2 signal wire from the SDU410 unit and disconnect the C4 connector.		
Action	Specification/Repair	Next Step

Check the signal wire for short to ground. - Place one test lead on the engine speed 2 signal wire at the SDU410 unit. - Place the other test lead to panel ground. - Place one test lead on the engine speed 1 signal pin at the C4 connector. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3A

STEP 3. Check the OEM wiring harness

STEP 3A. Check the engine speed 1 signal and return wires for an open at the C4 and C11 connectors.

Conditions :

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 1 sensor connectors.

Action	Specification/Repair	Next Step

Check the signal and return wires for an open. • Place one test lead on the engine speed 1 signal pin at C4 connector. • Place the other test lead on the engine speed 1 signal pin at the C11 connector. • Place one test lead on the engine speed 1 return pin at the C4 connector. • Place the other test lead on the engine speed 1 return pin at the C11 connector. • Place one test lead on the engine speed 1 signal pin at the sensor connector. • Place the other test lead on the engine speed 1 signal pin at the C11 connector. • Place one test lead on the engine speed 1 return pin at the sensor connector. • Place the other test lead on the engine speed 1 return pin at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	3A-1 Repair complete
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STEP 3A-1. Check the engine speed 2 signal and return wires for an open at the C4 and C11 connectors.

Conditions :

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 2 sensor connectors.

Action**Specification/Repair****Next Step**

Check the signal and return wires for an open. • Place one test lead on the engine speed 2 signal pin at C4 connector. • Place the other test lead on the engine speed 2 signal pin at the C11 connector. • Place one test lead on the engine speed 2 return pin at the C4 connector. • Place the other test lead on the engine speed 2 return pin at the C11 connector. • Place one test lead on the engine speed 2 signal pin at the sensor connector. • Place the other test lead on the engine speed 2 signal pin at the C11 connector. • Place one test lead on the engine speed 2 return pin at the sensor connector. • Place the other test lead on the engine speed 2 return pin at the C11 connector. Refer to the circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	3B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 3B. Check the engine speed 1 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.

Conditions :

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 1 sensor connectors.

Action	Specification/Repair	Next Step
Check the signal and return wires for a wire-to-wire short. • Place one test lead on the engine speed 1 signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 1 return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 1 signal pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. • Place one test lead on the engine speed 1 return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	3B-1

STEP 3B-1. Check the engine speed 2 signal and return wires for a wire-to-wire short at the C4 and C11 connectors.

Conditions : • Open the customer interface box. • Disconnect the C4, C11, and engine speed 2 sensor connectors.		
Action	Specification/Repair	Next Step

Check the signal and return wires for a wire-to-wire short. • Place one test lead on the engine speed 2 signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 2 return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the engine speed 2 signal pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. • Place one test lead on the engine speed 2 return pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3C

STEP 3C. Check the engine speed 1 signal wire for a short to ground at the C4 and C11 connectors.

Conditions :

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 1 sensor connectors.

Action	Specification/Repair	Next Step
Check the signal wire for short to ground. • Place one test lead on the engine speed 1 signal pin at the C4 connector. • Place the other test lead to panel ground. • Place one test lead on the engine speed 1 signal pin at the C11 connector. • Place the other test lead on panel ground. • Place one test lead on the engine speed 1 signal pin at the engine speed sensor 1 connector. • Place the other test lead on panel ground.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	3C-1

STEP 3C-1. Check the engine speed 2 signal wire for a short to ground at the C4 and C11 connectors.**Conditions :**

- Open the customer interface box.
- Disconnect the C4, C11, and engine speed 2 sensor connectors.

Action	Specification/Repair	Next Step
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Check the signal wire for short to ground. • Place one test lead on the engine speed 2 signal pin at the C4 connector. • Place the other test lead to panel ground. • Place one test lead on the engine speed 2 signal pin at the C11 connector. • Place the other test lead on panel ground. • Place one test lead on the engine speed 2 signal pin at the engine speed sensor 1 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO Repair : The troubleshooting steps must be checked again from the beginning. A fault mode should have been detected.	1A